

ANGERS AP, France

WMO: 073901

Lat: 47.5619N

Lon: 0.3169W

Elev: 194

StdP: 14.59

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.9	26.2	17.5	13.4	25.9	21.2	16.0	28.4	23.8	51.5	21.0	49.9	3.4	60	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	88.2	68.6	84.3	66.8	80.8	65.6	70.5	84.0	68.8	80.7	67.1	77.7	7.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.3	97.6	74.1	64.6	91.7	72.8	62.9	86.4	71.1	34.6	84.2	33.0	80.9	31.6	77.6	80.6

Extreme Annual Design Conditions

Extreme Annual WS							Extreme Annual Temperature		n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
					Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)			(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.2	17.4	15.2	DB		18.4	95.6	4.2	4.3	15.4	98.7	13.0	101.2	10.7	103.6	7.6	106.7		
			WB		17.7	74.2	4.3	2.1	14.5	75.7	12.0	77.0	9.6	78.2	6.5	79.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	40.6	41.9	46.4	51.6	56.7	63.4	67.2	65.4	61.2	55.0	47.0	41.8	(6)
	DBStd	11.19	7.02	6.88	5.69	6.01	5.80	6.58	5.95	5.54	5.43	6.47	6.49	6.75	(7)
	HDD50	1153	299	232	142	49	9	1	0	0	1	27	132	261	(8)
	HDD65	4566	758	647	577	403	265	104	43	60	137	313	541	718	(9)
	CDD50	2341	6	5	30	97	215	404	534	479	338	183	41	9	(10)
	CDD65	277	0	0	0	0	7	56	112	74	24	4	0	0	(11)
	CDH74	3101	0	0	0	35	103	649	1210	811	270	23	0	0	(12)
	CDH80	1004	0	0	0	1	13	221	433	276	56	4	0	0	(13)
	WSAvg	6.6	7.1	7.5	7.8	6.7	6.7	6.3	6.3	5.7	5.7	5.9	6.8	7.2	(14)
	PrecAvg	26.7	2.7	2.1	1.9	2.2	2.4	1.7	1.8	1.8	2.1	2.8	2.8	2.7	(15)
(16) Precipitation	PrecMax	37.8	5.5	4.1	5.3	6.2	3.8	4.1	5.0	4.3	5.1	7.7	5.3	5.8	(16)
	PrecMin	18.7	0.2	0.1	0.2	0.4	1.0	0.2	0.3	0.4	0.0	0.7	0.7	0.5	(17)
	PrecStd	4.7	1.3	1.1	1.2	1.4	0.8	1.0	1.2	1.2	1.5	1.4	1.2	1.3	(18)
	PrecStd	4.7	1.3	1.1	1.2	1.4	0.8	1.0	1.2	1.2	1.5	1.4	1.2	1.3	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	61.6	69.7	78.4	81.0	94.8	95.2	93.2	85.8	76.7	64.7	58.1	(19)
		MCWB	54.2	51.6	54.6	60.4	65.8	71.1	70.5	69.5	65.2	64.0	58.4	54.3	(20)
	2%	DB	55.3	56.8	64.4	73.6	76.9	85.9	89.3	87.4	80.8	71.6	61.2	56.8	(21)
		MCWB	53.1	50.9	52.6	58.7	63.0	67.8	69.2	68.5	64.7	62.3	57.5	53.8	(22)
	5%	DB	53.6	53.8	60.6	68.3	73.2	80.8	84.4	82.3	76.9	67.9	59.2	55.0	(23)
		MCWB	51.2	49.6	51.7	56.6	61.3	65.9	66.8	66.4	63.8	60.8	56.0	52.4	(24)
	10%	DB	51.6	51.9	56.9	64.2	69.5	76.9	80.7	78.0	73.2	64.7	57.2	53.2	(25)
		MCWB	49.1	48.3	50.4	54.0	59.6	64.2	65.8	64.2	62.1	59.4	54.1	50.7	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.3	53.8	56.5	63.1	68.5	74.4	72.8	72.4	69.3	66.2	60.3	56.5	(27)
		MCDB	56.2	57.9	64.3	72.9	79.0	90.5	89.9	88.5	78.1	71.7	62.6	57.4	(28)
	2%	WB	53.7	52.2	54.4	60.0	65.1	69.6	70.4	69.9	67.1	63.7	58.4	54.6	(29)
		MCDB	55.3	55.8	60.7	70.7	73.8	81.5	84.9	83.4	75.9	68.8	61.1	56.0	(30)
	5%	WB	51.7	50.7	52.9	57.6	62.4	67.2	68.6	67.9	65.3	62.0	56.5	52.7	(31)
		MCDB	53.7	53.5	58.8	67.6	70.7	78.7	81.3	78.1	73.8	66.5	59.4	54.6	(32)
	10%	WB	49.5	49.1	51.3	55.1	60.1	65.3	66.9	66.0	63.4	60.3	54.6	50.8	(33)
		MCDB	50.9	51.5	56.2	62.8	67.3	75.3	78.3	75.0	71.4	64.3	57.2	52.8	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	11.4	14.7	17.4	20.5	20.7	21.4	22.3	22.5	21.3	17.2	13.7	12.8	(35)
		MCDBR	11.3	17.0	23.7	26.9	27.2	27.3	28.0	30.4	26.9	19.4	14.5	13.1	(36)
		MCWBR	9.3	12.3	14.0	14.1	14.4	12.6	11.4	13.5	12.8	11.4	10.3	10.9	(37)
	5% WB	MCDBR	10.8	15.5	19.3	25.6	23.5	24.7	25.4	25.8	22.8	16.5	13.4	12.6	(38)
		MCWBR	9.7	12.0	12.4	14.3	14.2	12.6	11.5	13.4	12.9	10.8	10.4	11.4	(39)
(40) Clear-Sky Solar Irradiance	taub		0.326	0.344	0.373	0.395	0.399	0.408	0.402	0.390	0.367	0.362	0.335	0.321	(40)
	taud		2.439	2.384	2.300	2.242	2.279	2.287	2.305	2.356	2.399	2.419	2.439	2.415	(41)
	Ebn at Noon		238	258	267	272	274	272	272	270	266	248	231	224	(42)
	Edh at Noon		22	28	35	41	41	41	39	36	31	26	21	20	(43)
(44) All-Sky Solar Radiation	RadAvg		334	608	967	1417	1676	1868	1784	1579	1243	723	414	290	(44)
	RadStd		31	72	111	162	134	177	108	139	92	56	38	35	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page